
SnagIt

The Complete Guide

Every feature, every flag, and step-by-step scenarios for downloading, searching, watching, and the web interface.

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1. What Snagit is

Snagit is a single-file YouTube downloader written in Python. You give it a link and it saves a video or an MP3. You can also search YouTube from your terminal, play results in VLC or mpv, or run a local web interface that you can open from your phone.

It is a thin, friendly layer over **yt-dlp** and **FFmpeg**. It handles the awkward parts for you: the quality choice, playlist folders, progress display, and checking that your setup is healthy.

The four ways to use it

Mode	Start with	Best for
Interactive menu	<code>snagit</code>	You don't want to remember flags. It asks questions.
One-liner	<code>snagit URL -q 720</code>	Quick downloads, scripts, muscle memory.
Search & watch	<code>snagit -s TEXT</code>	Finding something, then playing or grabbing it without a browser.
Web interface	<code>snagit --web</code>	Point-and-click, queues, a home music player, and phone access.

Feature summary

- Download a single video at 360, 480, 720 or 1080p (default 720p).
- Download a whole playlist, or only chosen items (for example 1-5,8,10-12).
- Convert any video or playlist to MP3 (192 kbps).
- Search YouTube in a full-screen terminal list with thumbnails, then play or download.
- Audio-only playback for listening without the video.
- Web interface with search, playlist checklists, a live queue, a file browser and a music player.
- LAN mode so a phone on the same Wi-Fi can send links to your computer and stream your music.
- Built-in requirements checker with one-click Update yt-dlp, Install Deno and a Test button.
- Clean progress output by default, full yt-dlp output with **-v**.
- Works on Windows, Linux and macOS.

2. Setup

What you need

Item	Required?	Install / notes
yt-dlp (with yt-dlp-ejs)	Yes	<code>pip install -U "yt-dlp[default]"</code> The [default] extra includes the YouTube challenge solver.
FFmpeg	Yes	Needed to merge video and audio, and to make MP3s. Must be on your PATH.
JS runtime	Yes*	deno (recommended), node or bun. YouTube needs one or some formats go missing. *SnagIt can offer to install Deno.
VLC or mpv	For search	Also works with IINA and ffplay. VLC and mpv give up to 1080p.
Pillow	Optional	<code>pip install Pillow</code> Thumbnails in the search list.
qrcode	Optional	<code>pip install qrcode</code> Shows a QR code in <code>--lan</code> mode.

Run it

```
python snagit.py           # interactive menu
python snagit.py "URL" -q 720 # one-liner
python snagit.py --web      # web interface
```

Install as a command (macOS / Linux)

This lets you type **snagit** from anywhere instead of **python snagit.py**.

```
git clone https://github.com/Echo4-4/snagit.git
cd snagit
chmod +x snagit.py
mkdir -p ~/.local/bin
cp snagit.py ~/.local/bin/snagit
echo 'export PATH="$HOME/.local/bin:$PATH"' >> ~/.zshrc
source ~/.zshrc
snagit -h
```

Using bash? Replace `~/.zshrc` with `~/.bashrc`. To update later: run **git pull**, then copy the file over again.

The requirements checklist

Every time SnagIt starts it checks your setup. In the terminal, installed items show **[ALRIGHT]** and missing ones show **[JACK]** with a fix hint. In one-liner mode only warnings are printed, for example when FFmpeg is missing or the JS runtime is not found. The web interface shows the same list, with versions, on its Requirements page.

TIP

Run **snagit --web --check** to open straight to the Requirements page. It works even if yt-dlp is not installed yet, so you can install it from the browser.

Where files and settings go

What	Location
Downloads	~/Downloads/Snagit by default (change with -o DIR)
Single video	Title.mp4 straight in the output folder
Playlist	Playlist Title/001 - Video Title.mp4 in a folder named after the playlist
Config (player, web settings)	Windows: %APPDATA%\snagit\config.json Linux/macOS: ~/config/snagit/config.json

3. Interactive menu

Run **snagit** with no arguments (or add **-i** to force it). You see a banner, the requirements checklist, and then a short set of questions.

Step by step

- 1 Choose an output style: **Clean** (progress and video names only) or **Verbose** (full yt-dlp messages). Skip this question by starting with **-v**.
- 2 Choose what to snag: **Playlist**, **Single video**, **MP3 (audio only)**, or **Search YouTube & watch**.
- 3 Paste the link. For a playlist or single video, choose a quality (360, 480, 720 or 1080). For a playlist you can also enter a range such as 1-5,8,10-12, or leave it blank for everything.
- 4 For MP3 you first choose **Single video** or **Playlist**.
- 5 When it finishes, Snagit asks **Snag something else? (y/n)** so you can keep going in one session.

NOTE

Press Ctrl+C during a download to cancel just that job. In the menu you are returned to the prompt, not kicked out of Snagit.

4. Command line downloads

Give Snagit a link that starts with **http** and it runs straight away.

Examples

```
snagit URL -q 720           single video at 720p
snagit URL -p -q 1080 -r 1-5 first 5 videos of a playlist at 1080p
snagit URL -a               single video as MP3
snagit URL -a -p            whole playlist as MP3
snagit URL -p -v            playlist with full yt-dlp output
snagit URL -o ~/Music/yt    save somewhere else
```

All flags

Flag	Meaning	Notes
<code>-q 360 480 720 1080</code>	Video quality	Default 720. Ignored for MP3.
<code>-p, --playlist</code>	Treat the link as a playlist	Without it, only the single video is taken, even from a playlist link.
<code>-a, --audio</code>	MP3 (192 kbps)	With -s it means play audio only.
<code>-r RANGE</code>	Playlist items	e.g. 1-5,8,10-12. Needs -p ; otherwise ignored with a note.
<code>-o DIR</code>	Output folder	Also applies to --web .

Flag	Meaning	Notes
<code>-v</code>	Full yt-dlp output	Default is a clean progress bar.
<code>-s TEXT</code>	Search YouTube	Multiple words are fine, no quotes needed.
<code>-f, --first</code>	Skip the list, play top result	Needs -s .
<code>-P NAME</code>	Player	vlc, mpv, iina, ffplay, or a full path.
<code>-T, --no-thumbs</code>	No thumbnails	Search list is text only.
<code>-i</code>	Force interactive menu	
<code>--web</code>	Start web interface	Can't be combined with a link, -s or -i .
<code>--port N</code>	Web port	Default 8642. Needs --web .
<code>--no-open</code>	Don't open the browser	Needs --web .
<code>--check</code>	Open on Requirements page	Needs --web .
<code>--lan</code>	Allow other devices on the Wi-Fi	Needs --web .
<code>--lan-timeout M</code>	Idle minutes before LAN turns off	Default 30, 0 = never. Needs --lan .
<code>-V</code>	Show the version	

What the quality setting really does

SnagIt does not hard-cap the height. It sorts formats by how close they are to the quality you chose. If a video does not have exactly 720p, you get the nearest one instead of an error. Vertical videos such as Shorts are handled correctly. Video and audio are merged into an **.mp4** by FFmpeg.

Reliability details

- Private or deleted videos in a playlist are skipped, not fatal.
- Network hiccups are retried up to 5 times, including per-fragment retries.
- Up to 4 fragments download at once for speed.
- File names are made safe for Windows.
- A single Ctrl+C prints Cancelled. or Bye. and exits cleanly (exit code 130 in one-liner mode).

5. Search & watch (terminal)

Search YouTube without leaving your terminal. Results open full screen like **less**, with a small thumbnail next to each title, channel, length and view count. Scroll down and more results load automatically. Press Enter and the video streams online in your player, with no download.

Starting a search

```
snagit -s lofi hip hop      search and browse
snagit -s lofi -P mpv -q 1080 use mpv at 1080p
snagit -s lofi -T           no thumbnails
```

Listening without the video

```
snagit -s daft punk one more time -a -f  play the top result's sound right now
snagit -s daft punk -a                   browse, Enter plays just the audio
```

With **-f** the song plays in your terminal session and stops when you press Ctrl+C. In the list, Enter plays one song at a time (choosing another replaces it), **s** stops it, and quitting the list stops the music too.

Keys in the results list

Key	Action
Up / Down (or k / j)	Move through the list
PgUp / PgDn (or b / Space / f)	Page up / down
Home / End (or g / G)	Jump to first / last loaded result
Enter	Play the selected video (or only its audio in audio mode)
a	Switch between video and audio-only
s	Stop the audio started from the list
p	Switch player (your choice is remembered)
u	Change watch quality (360 / 480 / 720 / 1080)
d	Download the selected video (asks for quality)
m	Download the selected video as MP3
/	New search
r	Retry if loading results failed
q or Esc	Quit

Players

Snagit looks for VLC, mpv, IINA and ffplay, including VLC's default install folders on Windows and macOS. Use **-P** to choose one or to point at any other player.

```
snagit -s TEXT -P vlc
snagit -s TEXT -P "C:\path\to\player.exe"
```

VLC and mpv receive separate video and audio streams, so they can play up to 1080p. Other players get a single combined stream, usually 720p or lower. Your last player choice is saved in the config file.

Thumbnails

Thumbnails are drawn with coloured block characters, so they work in any colour terminal with no extra tools. Truecolor terminals (Windows Terminal, iTerm2, most Linux terminals) look best; macOS Terminal.app only has 256 colours, so it looks rougher. Without Pillow, with **-T**, or when **NO_COLOR** is set, the list is text only.

NOTE

Search mode needs an interactive terminal. It will not work when output is piped or run from a non-interactive script.

6. The web interface (--web)

A small web app served from your own computer. It uses only Python's standard library, needs no build step, and loads nothing from the internet apart from YouTube thumbnails. Downloads still run in your terminal, so you see the normal progress bar.

Starting it

```
python snagit.py --web
python snagit.py --web --port 9000 --no-open -o ~/Videos/snag
```

The terminal prints a link like **http://localhost:8642/?t=<token>** and your browser opens it. The token is random on every run, so always use the link from the terminal. The page remembers it in a cookie after the first visit. Without **--lan** the server only listens on your own computer (127.0.0.1).

The one box

The Home page has a single input. What you put in decides what happens:

You enter...	You get...
A playlist link (contains list= or /playlist)	A checklist of the playlist (loaded 100 videos at a time)
A single video link	One video card
Anything else	A YouTube search, 10 results at a time, with "Load more"

You can also press Ctrl+V anywhere on the page, click Paste, or drag a link onto the page.

Playlist checklist

- Tick videos with the checkboxes. Shift-click selects a range.
- **Select all** and **Select none** work on the whole playlist, even beyond the 100 loaded.
- Type a range like **1-5,8,10-12** and press **Tick range**.
- Pick a quality or MP3 above the list, then press **Download**.

Video and search cards

Each card offers some or all of these buttons:

Button	What it does
Watch	Plays the video, in the page or in VLC/mpv depending on Settings
Audio	Plays just the sound
Download	Queues a video download using the quality picker on Home
MP3	Queues an MP3 download
+ Queue	Adds the item to the Next up music list at the bottom of the page

Button	What it does
Next	Puts the item at the front of the Next up list

The pages

Page	What it does
Home	The one box, search results, playlist checklist and video cards.
Queue	Live progress for every job: current video, percent, speed, ETA, done or failed. Jobs that have not started can be removed.
Downloads	Finished files from your output folder. Play them in the page, or press Save for a copy. Filter box, "Play all audio", and seeking works.
Requirements	The [ALRIGHT]/[JACK] list with versions. Buttons: Update yt-dlp and Install Deno (both run in the terminal) and Test , which resolves a short video and reports the result.
Settings	Choose whether Watch and Audio play in the browser or in VLC/mpv. Saved to your config file.

Watch and Audio modes

- **Browser mode** plays inside the page using a combined stream, usually up to 720p.
- **Desktop mode** opens VLC or mpv on the computer. It cannot be started from a phone.
- **Next up** is a music list at the bottom of the page. It plays automatically, and Skip moves to the next song.
- **Music player**: on the Downloads page, use Play on a file, or Play all audio, to turn the page into a home music player.

The queue

Jobs run one at a time in a single worker, using the normal download code, so the terminal shows exactly what you would see in one-liner mode. Extra clicks join the queue. Closing the browser tab does not stop anything.

Stopping

Situation	Press Ctrl+C...
A download is running	Once: cancels only that job. SnagIt keeps running.
...and you press it again within 2 seconds	SnagIt quits.
Nothing is running	Once: SnagIt quits.

7. Phone and other computers (--lan)

LAN mode lets a phone, tablet or second computer on the same Wi-Fi use the web interface. The downloads still happen on your computer, and files stay there unless you press Save.

```
python snagit.py --web --lan
python snagit.py --web --lan --lan-timeout 60    # auto-off after 60 idle minutes
```

Pairing a phone

- 1 The terminal prints a warning that anyone on the network can reach the page, then the address, a 6-digit PIN, and a QR code. The QR code only appears if you ran **pip install qrcode**; otherwise type the printed address.
- 2 On the phone, scan the QR code or open the address.
- 3 Enter the PIN once.
- 4 The terminal asks **<ip> wants to connect. Allow? [y/N]**. Type **y**.
- 5 The phone is now paired for 30 days.

What a phone can do

- **Share to computer:** copy a YouTube link on the phone, paste it in the one box, and the download runs on the computer.
- **Save to phone:** on the Downloads page, press Save on a finished file.
- **Stream music:** on the Downloads page, press Play, or use Play all audio.
- A phone **cannot** launch a desktop player, run Update yt-dlp or Install Deno, or change Settings. Those work only on the computer itself.

Layers of protection

Layer	How it works
Token	A long random token in the link, new on every run.
PIN	6 digits, entered once per device. Wrong guesses are rate-limited, and after several the PIN changes and the terminal prints the new one.
Your approval	You must type y in the terminal for each new device.
Host checking	Only your computer's own names and addresses are accepted, which blocks DNS-rebinding tricks.
Idle timeout	After the idle time (default 30 min), LAN access switches off and all paired devices are cleared.

Turning LAN off

There is no separate off command. Either quit SnagIt with Ctrl+C, or let the idle timeout do it. Restart with **--lan** to turn it back on. The computer's own browser keeps working after a timeout. Using **--lan-timeout 0** disables the auto-off, so avoid it if you want it to shut itself down.

NOTE

Only use LAN mode on a network you trust, and quit SnagIt when you are done. The terminal deliberately prints a warning each time.

Firewall

System	What to do
Windows	Choose Allow when the firewall asks about Python.
Linux	<code>sudo ufw allow 8642</code> (or your --port).
macOS	Choose Allow when asked about incoming connections.

8. Scenarios

Real situations, and exactly what to do. Replace **URL** with your link.

Scenario 1: Save one video to watch offline

SITUATION

You are about to travel and want a tutorial on your laptop in good quality.

DO THIS

```
snagit "URL" -q 1080
```

WHAT YOU GET

A single **Title.mp4** in ~/Downloads/Snagit. If the video has no exact 1080p, you get the closest quality.

TIP

Add **-o ~/Desktop** to drop it somewhere easier to find.

Scenario 2: Turn a playlist into MP3s for a road trip

SITUATION

You have a playlist of songs and want them on your phone.

DO THIS

```
snagit "PLAYLIST_URL" -a -p
```

WHAT YOU GET

A folder named after the playlist with files like **001 - Song.mp3**, in playlist order at 192 kbps.

TIP

Remember **-p**. Without it only one video is taken, even if the link is a playlist.

Scenario 3: Only some videos from a long playlist

SITUATION

A 60-video course playlist, and you only need lessons 1-5, 8 and 10-12 at 720p.

DO THIS

```
snagit "PLAYLIST_URL" -p -r 1-5,8,10-12 -q 720
```

WHAT YOU GET

Just those items, numbered by their position in the playlist.

TIP

In the web interface do the same thing by typing **1-5,8,10-12** into the range box and pressing Tick range.

Scenario 4: Play a song right now, no browser

SITUATION

You want music on while you work and you already know the song name.

DO THIS

```
snagit -s daft punk one more time -a -f
```

WHAT YOU GET

Snagit finds the top result and plays only the audio in your player. Press Ctrl+C in the terminal to stop.

Scenario 5: Search, preview, then download the one you like

SITUATION

You are not sure which upload is the right one (live, remix, official).

DO THIS

- 1 Run `snagit -s artist song name`.
- 2 Move through the list with the arrow keys and press **Enter** to preview a result.
- 3 Press **a** to switch to audio-only if you just want to hear it.
- 4 When you find the right one press **d** for video (pick a quality) or **m** for MP3.
- 5 After the download, press Enter to return to the list. Press **/** to search something else, or **q** to quit.

WHAT YOU GET

The chosen file in your output folder, and the list waiting for your next search.

Scenario 6: Use a specific player at a specific quality

SITUATION

VLC opens by default, but you prefer mpv, and you want sharper video.

DO THIS

```
snagit -s conference talk name -P mpv -q 1080
```

WHAT YOU GET

Results open in mpv at up to 1080p. Your choice is remembered next time; press **p** in the list to switch.

Scenario 7: Grab a playlist visually with the web interface

SITUATION

You would rather tick boxes than type ranges.

DO THIS

- 1 Run `snagit --web` and let the browser open.
- 2 Paste the playlist link into the one box.
- 3 Tick videos. Shift-click selects a range; Select all and Select none are at the top.
- 4 Choose a quality or MP3 above the list and press **Download**.
- 5 Watch progress on the Queue page, or in the terminal.

WHAT YOU GET

Jobs run one after another. You can close the tab and they carry on.

TIP

Use the Queue page to remove jobs that have not started yet.

Scenario 8: Build a home music player

SITUATION

You downloaded lots of MP3s and want to play them from any room.

DO THIS

- 1 Run `snagit --web` on the computer that holds the files.
- 2 Open the **Downloads** page.
- 3 Use the filter box to narrow the list, then press **Play** on a song, or press **Play all audio**.
- 4 For songs not yet downloaded, use **+ Queue** on a search result to add them to the Next up list. Skip moves on.

WHAT YOU GET

The page acts as a music player. Seeking works, and you can leave it running.

Scenario 9: Send a link from your phone to your computer

SITUATION

You find a video on your phone and want it saved on the computer without touching the computer.

DO THIS

- 1 On the computer run `snagit --web --lan`.
- 2 On the phone, scan the QR code (or type the printed address) and enter the PIN.
- 3 In the terminal type **y** when it asks to allow the phone.
- 4 From then on, copy a YouTube link on the phone and paste it in the one box. Choose Download or MP3.
- 5 The download runs on the computer. Quit Snagit when you are finished.

WHAT YOU GET

The file lands in the computer's output folder. The phone stays paired for 30 days unless the idle timeout clears it first.

TIP

Install `qrcode` (`pip install qrcode`) once so you can scan instead of typing the address.

Scenario 10: Stream your music to your phone

SITUATION

You are on the sofa with your phone and your music is on the computer.

DO THIS

- 1 Pair the phone as in Scenario 9.
- 2 Open the **Downloads** page on the phone.
- 3 Press **Play** on a track, or **Play all audio**.
- 4 Want a copy on the phone? Press **Save** on a finished file.

WHAT YOU GET

Music streams over your Wi-Fi from the computer.

TIP

Keep the computer awake and SnagIt running. If LAN times out from inactivity, restart with `--lan` and pair again.

Scenario 11: Something stopped working

SITUATION

Downloads that used to work now fail, or you get "Requested format is not available".

DO THIS

- 1 Run `snagit --web --check` to open the Requirements page.
- 2 Press **Test**. It resolves a short video and reports the result.
- 3 If anything shows **[JACK]**, use its button or follow its fix hint.
- 4 If yt-dlp is out of date press **Update yt-dlp** (it runs in the terminal). Then restart SnagIt.

WHAT YOU GET

Most breakages come from an old yt-dlp, or a missing JS runtime or FFmpeg. This finds which.

TIP

Without the terminal UI, run `pip install -U "yt-dlp[default]"` and try again.

Scenario 12: First-time setup on a fresh computer

SITUATION

You just installed Python and want to check that everything is ready.

DO THIS

- 1 Run `pip install -U "yt-dlp[default]"`.
- 2 Install FFmpeg and make sure it is on your PATH.
- 3 Run `snagit --web --check`. If no JS runtime is found, press **Install Deno**.
- 4 Optional: install VLC or mpv for search and `pip install Pillow` for thumbnails.
- 5 Press **Test**. When it succeeds, you are ready.

WHAT YOU GET

A green Requirements page and a working download.

Scenario 13: Two things at once on one machine

SITUATION

Snagit is already running and you get "Can't use port 8642".

DO THIS

```
python snagit.py --web --port 9000
```

WHAT YOU GET

A second instance on another port. Normally the message just means an old Snagit is still running in another terminal, so closing it also fixes the problem.

Scenario 14: Cancel a download you started by mistake

SITUATION

A 3-hour livestream started downloading.

DO THIS

- 1 Go to the terminal running Snagit and press **Ctrl+C** once. Only that job is cancelled.
- 2 Prefer not to quit? Just leave it. Snagit stays running and moves to the next job.
- 3 To quit completely press Ctrl+C again within 2 seconds, or once when idle.

WHAT YOU GET

The current job stops and the queue moves on.

9. Troubleshooting

Problem	Fix
"Requested format is not available"	Install a JS runtime (deno, node or bun) and yt-dlp-ejs: <code>pip install -U "yt-dlp[default]"</code> . Then update yt-dlp.
MP3 or merging fails	FFmpeg is missing or not on your PATH.
"Missing or wrong token"	Open the link printed in the terminal, not just localhost:8642. The token changes every run.
"Can't use port 8642"	Snagit is probably already running. Close it or use --port 9000 .
Phone can't reach the page	Same Wi-Fi? Firewall allows the port? LAN mode not timed out? Started with --lan ?
Videos won't play or download	Requirements page, then Test. Use the buttons or hints for anything marked [JACK], then restart Snagit.
"No media player found"	Install VLC or mpv, or point to one with -P /path/to/player .
No thumbnails	Install Pillow (<code>pip install Pillow</code>). They are also off with -T or when NO_COLOR is set.
Search mode says it needs a terminal	Run it in a real interactive terminal, not a pipe or background script.
"--range only applies with -p"	Add -p to treat the link as a playlist.
Error about --lan or --port	Those flags need --web ; --lan-timeout needs --lan .
Wrong PIN keeps failing	After several wrong guesses the PIN changes. Read the new one from the terminal.

10. Quick reference

Most-used commands

I want to...	Command
Use the menu	<code>snagit</code>
Download a video at 720p	<code>snagit URL</code>
Download a video at 1080p	<code>snagit URL -q 1080</code>
Get an MP3	<code>snagit URL -a</code>
Get a whole playlist as MP3	<code>snagit URL -a -p</code>
Get playlist items 1-5	<code>snagit URL -p -r 1-5</code>
Search and browse	<code>snagit -s words here</code>
Play top result's audio	<code>snagit -s words here -a -f</code>
Start the web interface	<code>snagit --web</code>

I want to...	Command
...and allow my phone	<code>snagit --web --lan</code>
Check my setup in the browser	<code>snagit --web --check</code>
Change output folder	<code>snagit URL -o DIR</code>
See all flags	<code>snagit -h</code>

Made by Echo404. Follow: [@JeffreyPeter_](#)